NAME

mbfnv2navlab − converts an MB-System fast navigation (fnv) text file to a Kongsberg Navlab smoothed navigation binary file.

VERSION

Version 5.0

SYNOPSIS

mbfnv2navlab [**−I***file*] [**−O***file*] [**−V**] [**−H**]
mbfnv2navlab [**−−input**=*file*] [**−−output**=*file*] [**−−verbose**] [**−−help**]

DESCRIPTION

mbfnv2navlab converts an MB-System fast navigation (fnv) ASCII text file to a Kongsberg Navlab smoothed navigation binary file (in the format of *navlab_smooth.bin*).

The fnv format is a tab-separated ASCII text file produced by MB-System programs such as **mbpreprocess**. Lines beginning with **#** are treated as comments and skipped. Each data line contains at least ten tab-separated fields:

```
field 1: date and time − "yyyy mm dd hh mm ss.ssssss" (space-delimited)
field 2: epoch_seconds − Unix epoch time with microsecond precision
field 3: longitude     − decimal degrees, positive east
field 4: latitude      − decimal degrees, positive north
field 5: heading       − degrees, 0−360, clockwise from north
field 6: speed         − km/hr
field 7: draft         − meters, positive down
field 8: roll          − degrees, positive starboard up
field 9: pitch         − degrees, positive bow up
field 10: heave        − meters, positive up (discarded)
```

Four optional fields defining the port and starboard edges of swath coverage may follow; they are accepted but ignored.

The output Kongsberg Navlab binary format contains records each consisting of 21 IEEE 754 double-precision (64-bit) floating-point values in little-endian byte order with no file header (record size: 168 bytes). The mapping from fnv columns to binary record fields is:

```
field  0: time_d    <- epoch_seconds
field  1: latitude  <- latitude
field  2: longitude <- longitude
field  3: depth     <- draft
field  4: 0.0       (not available in fnv)
field  5: 0.0       (not available in fnv)
field  6: heading   <- heading
field  7: roll      <- roll
field  8: pitch     <- pitch
field  9: 0.0       (vertical velocity, not in fnv)
field 10: speed_ms  <- speed / 3.6 (km/hr to m/s)
field 11: speed_ms  <- speed / 3.6 (copy of field 10)
fields 12−20: 0.0   (uncertainties and covariance, not in fnv)
```

Heave from the fnv file is read but discarded; the navlab binary format does not have an explicit heave field.

Speed is converted from km/hr to m/s. Both speed fields (10 and 11) in the output are set to the same value; in original navlab_smooth.bin files these two fields typically differ slightly because they represent estimates from different sensors or algorithms.

If no input file is specified, **mbfnv2navlab** reads from standard input. If no output file is specified, it writes to standard output. Bytes are automatically swapped on big-endian host architectures so that the output is always little-endian.

Lines with fewer than ten tab-separated fields and records with retrograde or non-positive timestamps are skipped with a warning when verbose mode is active, and silently otherwise.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−H

−−help
Prints the program name, usage, and a description of the fnv-to-navlab field mapping, then exits.

−V

−−verbose
Increases the verbosity of **mbfnv2navlab**. With this option the program prints the input and output file names to standard error when they are opened, reports any skipped lines with their line numbers and the reason for skipping, and prints a final summary of the number of lines read, records written, and lines skipped.

−I *file*

−−input=*file*
Sets the path of the input fnv file. If this option is not given, the program reads from standard input.

−O *file*

−−output=*file*
Sets the path of the output Kongsberg Navlab binary file. If this option is not given, the program writes to standard output.

FNV FORMAT

The fnv (fast navigation) format is an MB-System internal ASCII text format for time series navigation and attitude data. Its properties are:

- Lines beginning with **#** are comments and are ignored.
- Data lines are tab-separated.
- The first tab field contains date and time as six space-separated values: year, month, day, hour, minute, and decimal seconds.
- The second tab field is the same instant expressed as decimal Unix epoch seconds (seconds since 1970-01-01 00:00:00 UTC). This field is used as the time source by **mbfnv2navlab**.
- At least ten tab fields are required. Four additional optional fields give the port and starboard edges of swath coverage; these are accepted but ignored.

NAVLAB BINARY FORMAT

The navlab_smooth.bin format has the following properties:

- No file header of any kind.
- Each record is exactly 168 bytes: 21 values × 8 bytes (float64).
- All values are IEEE 754 double-precision floating point, little-endian byte order.
- Nominal sample rate of original Navlab output: 100 Hz (10 ms steps). Output from **mbfnv2navlab** will have the sample rate of the input fnv file, which is typically much lower.
- Fields not available from the fnv input are set to zero: fields 4, 5, 9, and 12 through 20. In original Navlab output these carry quantities such as position offsets, vertical velocity, and Kalman filter uncertainties that are not present in fnv files.

LIMITATIONS

Because the fnv format carries only a subset of the information in an original navlab_smooth.bin file, the output of **mbfnv2navlab** cannot replicate an original Navlab file exactly. The following fields are lost:

- **fields 4 and 5**: unknown quantities (possibly position offsets), set to 0.0.
- **field 9**: vertical velocity (m/s), set to 0.0.
- **fields 12−20**: speed components, Kalman filter uncertainties, and covariance cross-terms, all set to 0.0.
- **field 11**: a second speed estimate from an alternate sensor or algorithm; set equal to field 10 (the speed converted from the fnv km/hr value).

The intended use of **mbfnv2navlab** is to produce a navlab-format binary file from corrected or adjusted navigation computed by MB-System tools such as **mbnavadjust**, for subsequent use as a navigation source in **mbpreprocess** or **mbprocess** with navlab binary format (format id 12).

EXAMPLES

Convert an fnv file to a navlab binary file:

```
mbfnv2navlab −I mission.fnv −O adjusted_nav.bin
```

Use standard input and output (e.g. within a pipeline):

```
cat mission.fnv | mbfnv2navlab > adjusted_nav.bin
```

Convert with verbose progress reporting:

```
mbfnv2navlab −−verbose −−input=mission.fnv −−output=adjusted_nav.bin
```

A common workflow: generate a corrected fnv file from processed swath data, then use the resulting navlab binary file as the navigation source for subsequent processing:

```
# Generate fnv from existing processed data
mblist −I datalist.mb−1 −O tMXYHSDbRP > corrected.fnv
# Convert to navlab binary format
mbfnv2navlab −I corrected.fnv −O corrected_nav.bin
# Use as nav source in mbpreprocess
mbpreprocess −−format=261 −−input=datalist.mb−1 \
  −−nav−file=corrected_nav.bin −−nav−file−format=12 \
  −−sensordepth−file=corrected_nav.bin −−sensordepth−file−format=12 \
  −−attitude−file=corrected_nav.bin −−attitude−file−format=12 \
  −−heading−file=corrected_nav.bin −−heading−file−format=12
```

SEE ALSO

mbsystem(1), **mbnavlab2fnv**(1), **mbpreprocess**(1), **mbprocess**(1), **mbnavadjust**(1), **mblist**(1)

BUGS

Fields 4, 5, 9, and 12−20 of the navlab binary record are set to 0.0 in the output because the fnv format does not carry those quantities. If the resulting file is used as a navigation source in **mbpreprocess** or **mbprocess**, only fields 0−3 and 6−10 are used and the zeroed fields have no effect.

The sample rate of the output file is determined by the input fnv file and will generally differ from the 100 Hz rate of original Navlab output.